

CIVIC AND TRAFFIC AI COMPLAINT PRIORITY MANAGEMENT APPLICATION

A

PROJECT PROPOSAL REPORT

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Abstract

CIVICAI is a proposed AI-assisted mobile complaint management system designed to improve the way citizens report and authorities manage civic and traffic-related problems such as potholes, garbage, water issues, streetlight problems, illegal parking, helmet violations, and other traffic issues. The system will allow citizens to submit complaints using text, images, and location information, while an authority dashboard will provide a centralized platform for monitoring and managing complaints. Natural Language Processing (NLP) and Deep Learning techniques will be used to classify text and image-based complaints, while duplicate detection will identify repeated complaints. The system will combine the predicted base urgency with the number of duplicate reports to calculate the final priority and route complaints to the relevant department. Different machine learning and deep learning models will be compared using appropriate performance metrics, and the best-performing models will be integrated into the application. The expected outcome is a functional mobile-based complaint management system that can reduce manual complaint processing, improve prioritization and routing, identify repeated issues, and support authorities in responding to important civic and traffic problems more efficiently.

Keywords: CIVICAI, Complaint Management, Natural Language Processing, Machine Learning,

Duplicate Detection, Complaint Prioritization, Civic Issues, Traffic Issues.

Table of Contents

Introduction	1
Literature Review.....	2
Aims and Objectives	5
Aims	5
Objectives	5
Methodology.....	6
Description of Project Approach.....	6
Data Collection Methods	6
Tools, Technologies, or Materials Required.....	7
Timeline and Milestone	7
Expected Outcomes.....	9
Testing and Evaluation.....	10
Assessment of Objectives	10
Testing Strategy	10
Key Performance Indicators (KPIs)	10
Legal, Social, and Ethical Issues	12
Legal Issues.....	12
Social Issues.....	12
Ethical Issues	12
Guidelines to Minimize Issues.....	12
Resource Requirement	13
Hardware Requirements	13
Software Requirements	13
Dataset Requirement	13
Development Resources	13
References	14
Appendix	15

Introduction

Urban municipalities need effective ways to receive and manage complaints about public problems such as waste, damaged roads and other civic services. The Government of Nepal's Waste Management Baseline Survey of Nepal 2020 examined waste management across 293 urban municipalities, showing the scale of municipal service-management responsibilities in Nepal (Central Bureau of Statistics, 2021).

CIVICAI is proposed to transform the traditional complaint process into a mobile-based digital system. Instead of submitting complaints through written methods, citizens will be able to report civic and traffic problems through a mobile application. The proposed system will allow users to submit complaints such as potholes, garbage, water problems, streetlight issues, no-helmet violations, red-light violations, wrong-side driving and illegal parking. After submission, the AI component will assist in classifying the complaint, assigning a priority score and detecting potentially duplicate complaints. The processed complaints will then be provided through an authority dashboard for monitoring and management. The feasibility of using mobile applications and AI for civic complaint management has already been demonstrated by Shama et al. (2024), who developed a mobile complaint system using deep learning to automatically categorize municipal complaints and provide complaint information to authorities.

The significance of CIVICAI is that it combines citizen complaint reporting, AI-based classification, priority scoring, duplicate detection and authority-side management within one system. AI has already been explored in citizen complaint management, including deep-learning-based complaint categorization (Shama et al., 2024) and conversational AI for complaint management (Karren et al., 2024).

Literature Review

Rajapandian and Rudhrakumar (2026) proposed an AI-driven civic complaint management system with priority modelling to improve the handling of municipal complaints by addressing issues such as manual processing, incorrect routing, and delayed prioritization. The system allows citizens to submit complaints containing text, images, and location information, after which AI techniques are used to classify and process the complaints. BERT is used for complaint classification, ResNet is used to analyze the severity of complaint images, XGBoost is used for priority prediction, and Rasa is used for chatbot-based intent recognition. The system classifies complaints into categories such as sanitation, drainage, road damage, water supply, and streetlight issues and automatically routes them to the appropriate municipal departments. The authors reported accuracies of 92% for BERT-based complaint classification, 90% for ResNet-based image severity analysis, 88% for XGBoost-based priority prediction, and 91% for Rasa intent recognition, indicating that AI can improve complaint processing, routing, prioritization, and communication between citizens and municipal authorities. Therefore, this study provides a strong foundation for the proposed AI-Based Intelligent Civic Complaint Prioritization and Routing System, while the proposed work aims to extend these capabilities for a more comprehensive civic complaint management solution.

Suganthi et al. (2026) proposed an AI-Based Smart Civic Complaint Management System to improve the efficiency of municipal complaint handling by automating complaint classification, duplicate detection, prioritization, routing, and predictive analysis. The system allows citizens to submit complaints containing text descriptions, images, and location details, after which NLP techniques are applied to preprocess the textual information and TF-IDF with Logistic Regression or Naïve Bayes is used to classify complaints into categories such as garbage

management, water supply, road damage, and sanitation. For image-based complaints, a CNN-based image analysis module is incorporated to identify civic issues and verify complaint authenticity, while cosine similarity is used to detect duplicate complaints. The system also assigns complaints into high, medium, and low priority levels based on factors such as severity, complaint type, and location, and automatically routes them to the appropriate municipal department. The reported results show an improvement from 65% to 92% in accuracy, with precision, recall, and F1-score increasing from 0.60, 0.62, and 0.61 to 0.89, 0.90, and 0.89, respectively, demonstrating the potential of integrating AI, ML, NLP, and image processing for automated civic complaint management. However, although the study incorporates several components closely related to the proposed research, its approach primarily uses traditional text classification techniques such as TF-IDF with Logistic Regression/Naïve Bayes and describes the CNN, prioritization, and predictive modules at a relatively high level (Suganthi et al., 2026).

Dr. Basanta Joshi and colleagues explored Nepali grievance classification using DistilBERT, focusing on the automated classification of public complaints collected from platforms such as Hello Sarkar and Kathmandu Metropolitan City. The researchers compared conventional approaches such as Support Vector Machine (SVM) and Naïve Bayes with transformer-based models and applied Masked Language Modeling (MLM) for DistilBERT pre-training followed by transfer learning and fine-tuning for grievance classification. Using a dataset of 7,563 grievances across 13 categories, the study found that traditional SVM and Naive Bayes models achieved comparatively lower performance, while DistilBERT achieved an F1-score of approximately 0.87 when evaluated on five selected categories. Although NepaliBERT performed better than DistilBERT, the findings demonstrated that transformer-based models can

effectively capture contextual information in Nepali grievance text and provide a promising approach for automated complaint classification.

According to Dede et al. (2023), an AI-assisted mobile traffic surveillance system was developed to automatically detect six major traffic violations, including red-light violations, illegal use of breakdown lanes, insufficient following distance, pedestrian crossing violations, illegal parking, and parking on marked crosswalks. The proposed system integrates YOLOv5 for detecting vehicles, pedestrians, and traffic signs with strongSORT for tracking objects across consecutive video frames. In addition, WPOD-NET was employed for license plate detection, while image-processing techniques and MobileNet were used for license plate character recognition. The researchers utilized both online traffic videos and self-recorded dash-camera footage, along with additional traffic-sign datasets, to develop and evaluate the system. Their experimental results showed that the proposed algorithms detected 23 violating vehicles across 14 traffic scenes. The study demonstrates the potential of combining object detection, object tracking, and license plate recognition for automated and mobile-based traffic surveillance. However, the evaluation was conducted on a relatively limited number of scenes, indicating the need for larger and more diverse datasets and further validation before widespread real-world deployment. Overall, Dede et al. (2023) provided a useful foundation for developing AI-based traffic violation detection systems capable of supporting automated monitoring and reporting.

Aims and Objectives

Aims

To develop an AI-assisted mobile civic complaint management system that enables citizens to digitally report civic and traffic issues while supporting authorities through complaint classification, priority scoring, duplicate detection and centralized complaint management.

Objectives

- To develop a mobile application that enables citizens to submit civic and traffic complaints digitally.
- To implement an AI-based mechanism for automatically classifying submitted complaints.
- To develop a priority-scoring mechanism to help identify complaints requiring greater attention.
- To implement duplicate detection to identify similar or repeated complaints.
- To develop an authority dashboard for viewing, organizing and monitoring submitted complaints.

Methodology

Description of Project Approach

The project will follow a design and development approach. Citizens will submit civic or traffic complaints through text, images, and location. NLP will classify text complaints, deep learning will analyze images, and machine learning will determine complaint priority. Duplicate complaints will also be identified, and reports will be routed to the relevant authority. The system will first preprocess the submitted information before applying the appropriate AI model. The output from these models will be combined to categorize the complaint, determine its importance, and support the routing process. Finally, the processed complaints will be displayed on an authority dashboard for monitoring and management.

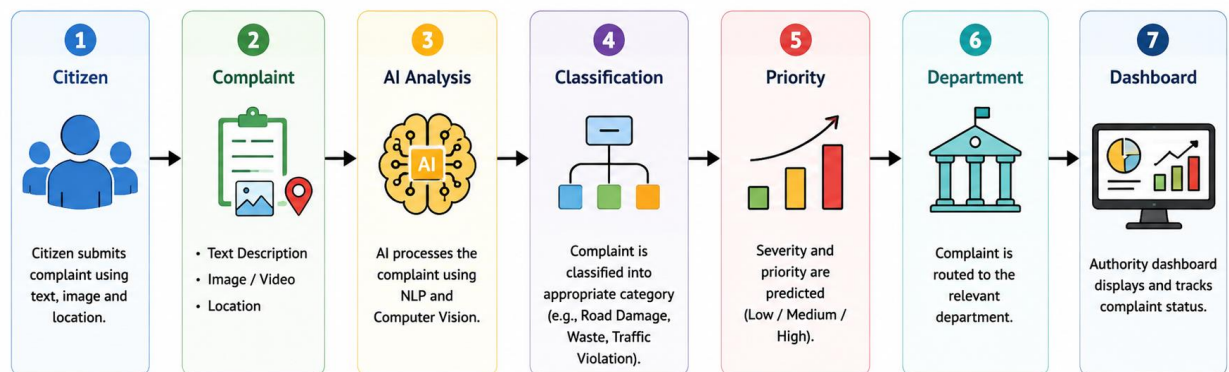


Figure 1: Project Workflow

Data Collection Methods

Data will be collected from public datasets and manually labelled data for civic issues such as potholes, waste, drainage, and streetlights, and selected traffic violations. Data will include complaint text, images, category, location, severity, and priority where available. Text will be cleaned and tokenized, while images will be resized and normalized before model training. The collected data will be organized according to the categories required by the proposed system.

Where sufficient Nepal-specific data is not available, suitable publicly available data and manually labelled examples will be considered. The prepared dataset will then be divided into training, validation, and testing data for developing and evaluating the AI models.

Tools, Technologies, or Materials Required

- Python: AI/ML development.
- Hugging Face Transformers: NLP models such as BERT/DistilBERT.
- Models: Logistic Regression/SVM, CNN, DenseNet/ResNet.
- FastAPI/Flask: Backend and API development.
- MySQL/PostgreSQL: Database management.
- Postman (API testing).
- JS, React Native, API Routes.
- Git/GitHub: Version control and project management.

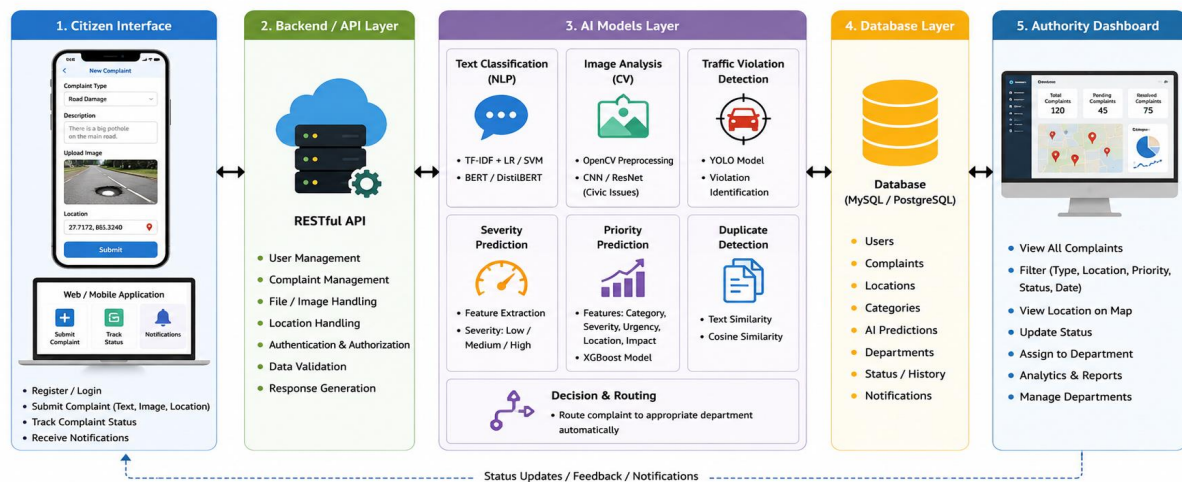


Figure 2: System Architecture

Timeline and Milestone

Phase	Activities

Month 1	Requirements analysis and system design
Month 2	Data collection, cleaning, and preprocessing
Month 2	AI model development and model selection
Month 3	Backend, database, and dashboard development
Month 4	AI integration, system testing, and improvements
Month 5	Final documentation, presentation, and project completion

Expected Outcomes

The project is expected to produce a functional mobile application, AI-based complaint-processing system, database and authority dashboard. Citizens will be able to submit civic and traffic complaints digitally, while the AI component will classify complaints, generate priority scores and identify potentially duplicate complaints. The authority dashboard will provide a centralized platform for authorized personnel to view and manage submitted complaints. These outcomes will address the limitations of a traditional complaint process by providing a more structured digital method for reporting and managing civic and traffic issues. The feasibility of mobile-based complaint reporting with AI-supported classification is supported by existing research (Shama et al., 2024).

Testing and Evaluation

The proposed system will be tested to ensure that it accurately classifies complaints, assigns appropriate priority levels, detects duplicate complaints, and functions correctly as a complete complaint management platform

Assessment of Objectives

- The project objectives will be considered achieved if the system can:
- Correctly classify civic and traffic complaints.
- Correctly predict the base urgency of complaints.
- Successfully identify duplicate complaints.
- Calculate the final priority based on urgency and duplicate count.
- Correctly route complaints to the appropriate department.
- Allow citizens and authorities to perform all intended functions without errors.

Testing Strategy

The following testing methods will be used:

- **Unit Testing:** Test individual modules such as login, complaint submission, image upload, AI prediction, and dashboard functions.
- **Integration Testing:** Verify that the frontend, backend, AI models, and database communicate correctly.
- **System Testing:** Test the complete system workflow from complaint submission to authority response.
- **Model Evaluation:** Evaluate machine learning models using unseen test data.

Key Performance Indicators (KPIs)

Component	Evaluation Metric
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Text Classification	Accuracy, Precision, Recall, F1-Score
Image Classification	Accuracy, Precision, Recall, F1-Score
Duplicate Detection	Similarity Accuracy, Precision, Recall
Overall System	Response Time, Processing Time, System Reliability
User Interface	Ease of Use
Complaint Processing	Correct Department Routing and Priority Assignment

The model with the best performance during comparison will be selected for deployment.

Legal, Social, and Ethical Issues

The proposed system involves handling citizen-generated complaints; therefore, legal, social, and ethical responsibilities must be considered throughout the project.

Legal Issues

- The project will use publicly available datasets or datasets created specifically for academic purposes.
- Copyright laws will be respected when collecting datasets, images, or other resources.
- The system will comply with applicable data protection and privacy regulations.

Social Issues

- The system aims to improve communication between citizens and government authorities.
- Faster complaint prioritization can improve public service delivery and increase citizen satisfaction.

Ethical Issues

- AI predictions should assist authorities rather than replace human decision-making.
- Citizens' personal information and uploaded images will remain confidential.

Guidelines to Minimize Issues

- Encrypt sensitive user information.
- Collect only the minimum information required.
- Use authorized and publicly available datasets.
- Test models before deployment.
- Maintain transparency regarding how AI prioritizes complaints.

Resource Requirement

The successful implementation of the project requires appropriate hardware, software, datasets, and development tools.

Hardware Requirements

- Laptop

Software Requirements

- Operating System: Windows 10/11
- Visual Studio Code

Dataset Requirement

Text Dataset: A dataset containing civic and traffic complaints with labels such as:

- Complaint Description
- Complaint Category
- Base Urgency
- Department

Image Dataset: A labelled image dataset containing:

- Potholes
- Garbage
- Broken Streetlights
- Illegal Parking

Development Resources

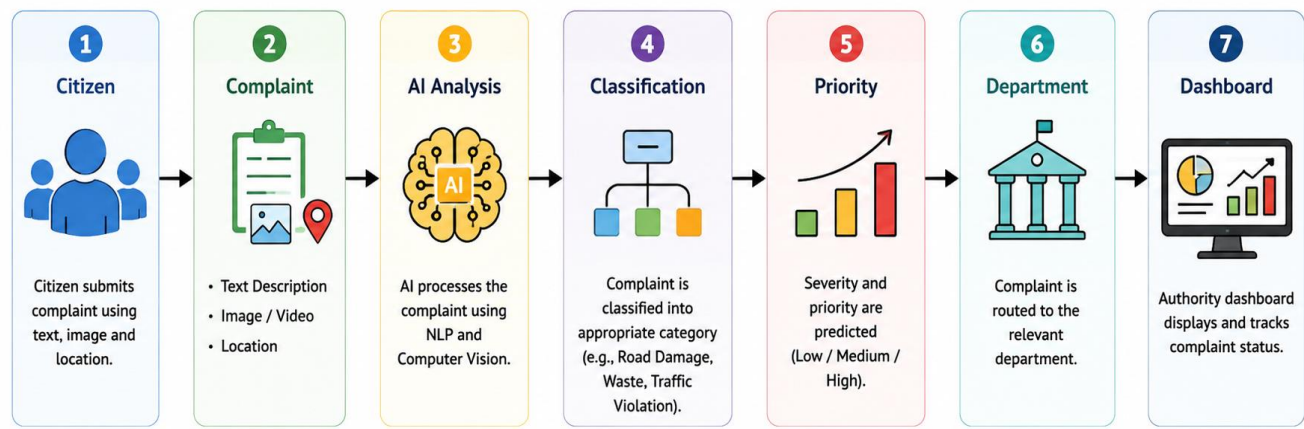
- Project Supervisor
- Research Papers and Journals

References

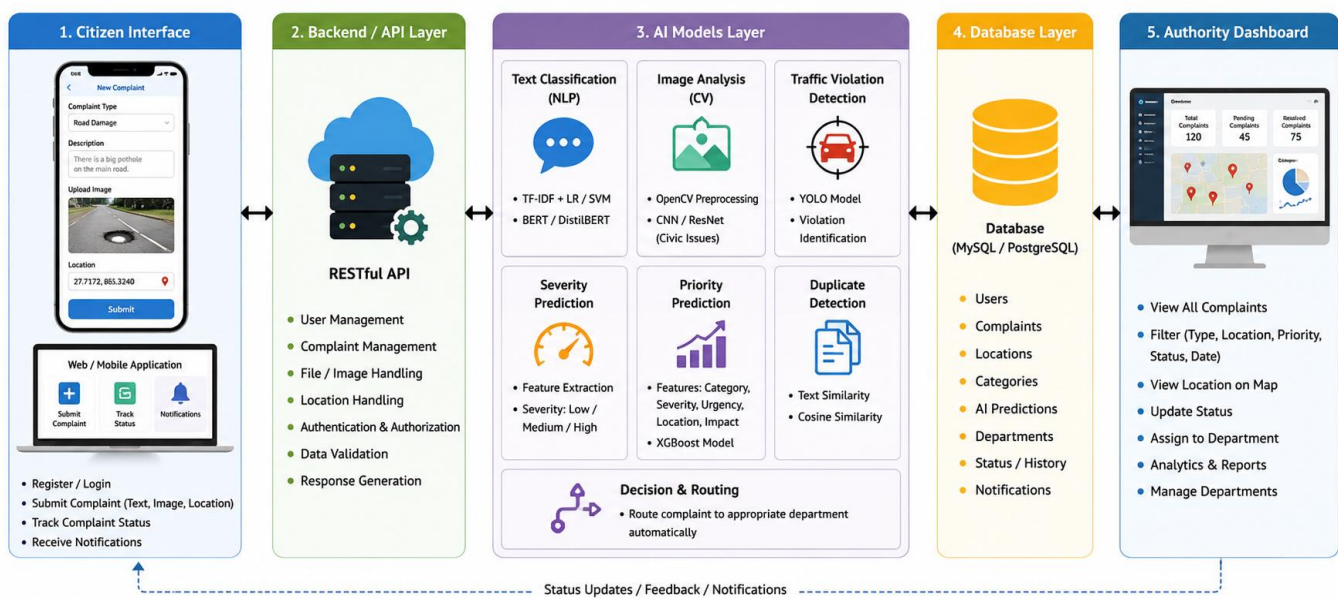
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Appendix

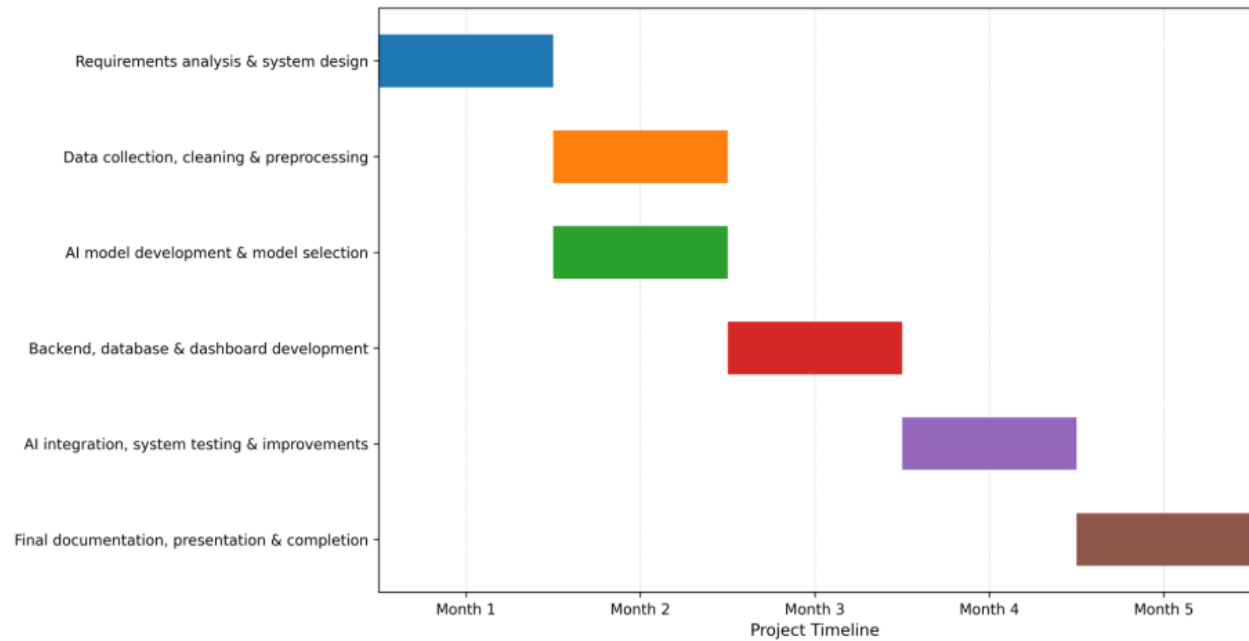
Appendix 1: Project Workflow



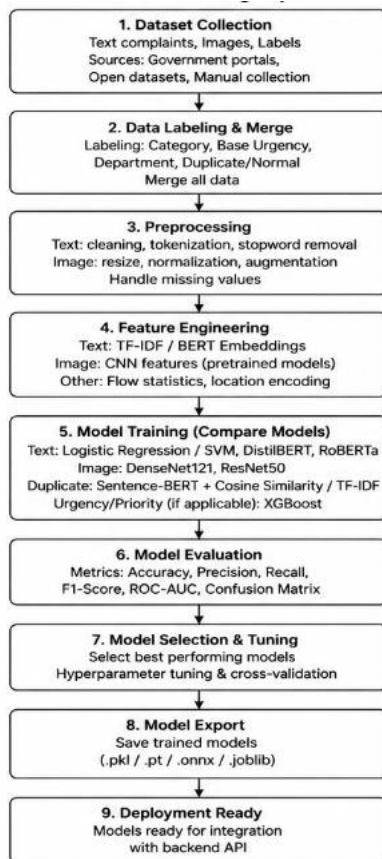
Appendix 2: System Architecture



Appendix 3: Gantt Chart



Appendix 4: AI/ML Training Pipeline



Appendix 5: Application Development Pipeline

